



NORTH SYDNEY BOYS HIGH SCHOOL

2008 YEAR 11 EXTENSION 1 ASSESSMENT TASK 3

Mathematics

General Instructions

- Working time – **50 minutes**
- Write on one side of the paper using blue or black pen
- This examination paper contains 5 sections
- All necessary working should be shown in every question
- Each new **section** is to be started on a **new page**.
- Attempt all questions

Total Marks (50)

Class Teacher:

(Please tick or highlight)

- ☐ Mr Barrett
- ☐ Mr Weiss
- ☐ Mr Lowe
- ☐ Mr Rezcallah
- ☐ Mr Taylor
- ☐ Ms Wei

Student Name:

No of working pages used: ____ (fill this at the end of the exam)

Section	Intro Calc	Plane Geom	Co-ord Geom	Logs and Indices	3D Trig	Total	Total
Mark	<u>11</u>	<u>11</u>	<u>13</u>	<u>10</u>	<u>5</u>	<u>50</u>	<u>100</u>

Question 1 Introductory Calculus

a) Differentiate using appropriate techniques:

(7 marks)

(i) $y = \frac{1}{3}x^3 + x - 1.5$

(iii) $y = x(3x - 2)^4$

(ii) $f(x) = -\sqrt[4]{x^3} + \frac{1}{x}$

(iv) $f(x) = \frac{2x^2}{3x-1}$

b) Evaluate : $\lim_{x \rightarrow \infty} \frac{2x}{x-1}$

(2 marks)

c) Find the derivative of $f(x) = 3x - 1$ from first principles.

(2 marks)

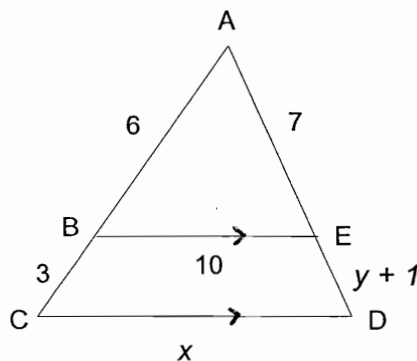
Question 2 Plane Geometry

a) Find the number of sides of a regular polygon has if each interior angle is 144° .

(2 marks)

b) Find the values of x and y . (No reasons required)

(4 marks)

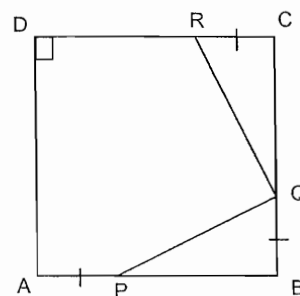


c) In this square ABCD, R, Q and P are points chosen such $AP = BQ = RC$. (Show all necessary working and reasoning)

(5 marks)

(i) Prove that $\triangle PBQ \cong \triangle QCR$

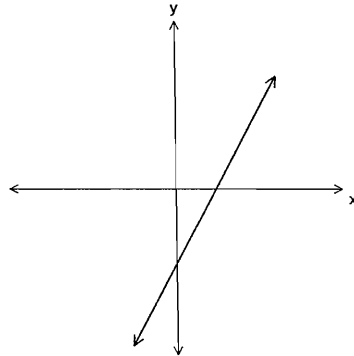
(ii) Prove that $\angle PQR$ is a right angle.



Question 3 Coordinate Geometry

- a) Find the coordinates of the point that divides AB, $A(3, -1)$ and $B(9, 7)$, *externally* in the ratio of $3 : 2$. (3 marks)

- b) In the diagram, The line L_1 passes through the point $(2,0)$ on the x-axis, and $(0, -6)$ on the y-axis.



- (i) Show that the equation of L_1 is $3x - y - 6 = 0$. (2 marks)

- (ii) The line L_2 has equation $2x + y - 14 = 0$. By using the 'k method' or otherwise, find the equation of a line that passes through the intersection of L_1 and L_2 and the point $(1, 4)$. (3 marks)

- (iii) Calculate the perpendicular distance from the point $(1, 4)$ to L_2 . (2 marks)

- c) Sketch the region satisfying: $y < \sqrt{64 - x^2}$ and $y \geq -4$ (3 marks)

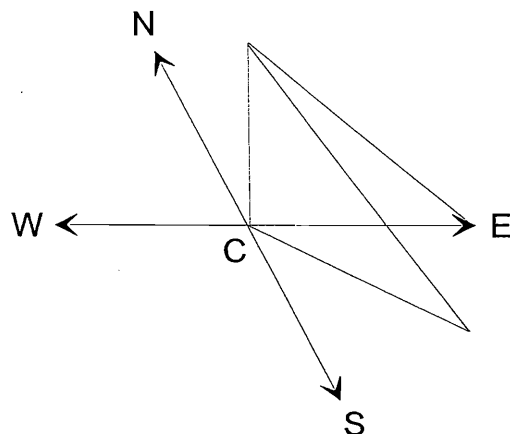
Question 4 Logarithm and Indices

- a) Calculate the value of $\log_3 15$, correct to 2 decimal places. (1 mark)
- b) If $\log_{64} a = \frac{1}{3}$, find the value of a . (1 mark)
- c) Give the exact value of $(\log_6 4 + \log_6 9)^2$ (2 marks)
- d) Simplify $b^{2 \log_b 7}$ (1 mark)
- e) Solve for x : $\log_3(x + 3) - \log_3(x - 3) = 1$ (3 marks)
- f) If $2^{x+1} < 12$, find the values of x to 2 decimal places. (2 marks)

Question 5 3D Trigonometry

Point A is located east of a hill and point B has a true bearing of 135° from the hill. The angle of elevation to the top of the hill from A is 43° , from B 62° . A and B lie on level ground and are 600m apart.

Let the base of the hill be point C, and its perpendicular height be h .



Copy the diagram in your answer booklet.

- (i) Show that BC has a distance of $\frac{h}{\tan 62^\circ}$ (1 mark)
- (ii) Derive a similar expression for AC . (1 mark)
- (iii) Hence, find the value of h to the nearest metre. (3 marks)

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Q1 Introductory Calculus

a i) $y' = x^2 + 1$

1

$$\text{ii) } f(x) = -x^{\frac{3}{4}} + x^{-1}$$

$$f'(x) = -\frac{3}{4}x^{-\frac{1}{4}} - x^{-2}$$

$\uparrow$ $\uparrow$
 1 1
 each

2

$$\text{iii) } u = x \quad v = (3x-2)^4$$

$$u' = 1 \quad v' = 4(3x-2)^3 \times 3$$

$$= 12(3x-2)^3$$

1

$$y' = (3x-2)^4 + 12x(3x-2)^3$$

$$(3x-2)^3 [3x-2 + 12x] = (3x-2)^3 (15x-2)$$

1

$$\text{iv) } u = 2x^2 \quad v = 3x-1$$

$$u' = 4x \quad v' = 3$$

1

$$f'(x) = \frac{4x(3x-1) - 6x^2}{(3x-1)^2}$$

(1)

$$= \frac{12x^2 - 4x - 6x^2}{(3x-1)^2}$$

$$= \frac{6x^2 - 4x}{(3x-1)^2}$$

2

Q1 - cont.

b) $\lim_{x \rightarrow \infty} \frac{2x}{x-1}$

$$= \lim_{x \rightarrow \infty} \frac{2x}{x}$$

1

$$\frac{x}{x} = \frac{1}{1}$$

$$= \lim_{x \rightarrow \infty} \frac{2}{1 - \frac{1}{x}}$$

answer only?

$$= \frac{2}{1-0}$$

$$= 2$$

1 ✓

c) $f(x) = 3x-1$

$$f(x+h) = 3(x+h)-1$$

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{3(x+h)-1 - (3x-1)}{h}$$

1

$$= \lim_{h \rightarrow 0} \frac{3x+3h-1-3x+1}{h}$$

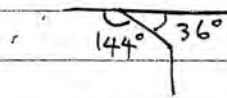
$$= \lim_{h \rightarrow 0} 3$$

$$= 3$$

1

Q2. Plane Geometry

a)



$$\text{ext } \angle = 36^\circ$$

$$\therefore \text{no. of sides} = \frac{360^\circ}{36^\circ}$$

$$= 10$$

1

1

$$\text{b) } \frac{6}{3} = \frac{7}{y+1}$$

1

$$\frac{1}{2} = \frac{y+1}{7}$$

$$\frac{7}{2} = y+1$$

$$y = \frac{7}{2} - 1$$

$$= \frac{5}{2} \text{ or } 2\frac{1}{2} \text{ or } 2.5$$

1

$$\frac{6}{6+3} = \frac{10}{x}$$

1

$$\frac{3}{2} = \frac{x}{10}$$

$$x = 15$$

1

c) (i) In ΔPBQ and ΔQCR

$$\bullet BQ = RC \text{ (given)}$$

$$\bullet \angle PBQ = \angle QCR = 90^\circ \text{ (}\angle\text{s in a square)}$$

$$\bullet \text{ since } AB = BC \text{ (= sides of square)}$$

$$\therefore AB - AP = BC - BQ$$

$$\therefore PB = QC$$

1

$$\therefore \Delta PBQ \equiv \Delta QCR \text{ (SAS)}$$

1

Q2 cont.

(ii) construct a line TQ , $\parallel$ to AB and CD

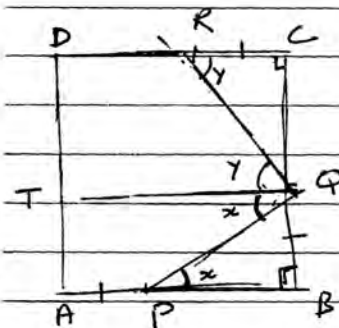
$$\angle TQP = \angle BPQ \text{ (alt } \angle\text{s, } TQ \parallel PB) = x$$

$$\angle CRQ = \angle RQT \text{ (alt } \angle\text{s, } DC \parallel TQ)$$

$$= \angle BQP \text{ (corresp. } \angle\text{s in } \triangle\text{s)}$$

$$\therefore \angle TQP + \angle RQT = \angle BPQ + \angle BQP$$

$$= 90^\circ \text{ (}\angle\text{sum, right } \triangle\text{)}$$



23 Coordinate Geometry

a) A(3, -1) B(9, 7)

$$3 : -2 \quad (\text{or } -3 : 2)$$

$$\begin{aligned} x &= \frac{3 \times 9 + (-2) \times 3}{3 + (-2)} && 1 \text{ for method} \\ &= \frac{27 - 6}{1} && 1 \text{ for correct } x \\ &= 21 && 1 \text{ for correct } y \end{aligned}$$

$$\begin{aligned} y &= \frac{3 \times 7 + (-2) \times (-1)}{3 + (-2)} \\ &= 21 + 2 \\ &= 23 \end{aligned}$$

b) i) gradient of $L_1 = \frac{-6 - 0}{0 - 2} = 3$ 1 for gradient

use either (2, 0) or (0, -6)

eg: $y - y_1 = m(x - x_1)$
 $y - 0 = 3(x - 2)$

$$y = 3x - 6$$

$$3x - y - 6 = 0 \quad 1 \text{ for correct answer}$$

ii) $(3x - y - 6) + k(2x + y - 14) = 0$ 1

sub in (1, 4)

$$(3 - 4 - 6) + k(2 + 4 - 14) = 0$$

$$-7 - 8k = 0$$

$$k = -\frac{7}{8} \quad 1 \text{ for } k \text{ value}$$

$$\therefore (3x - y - 6) - \frac{7}{8}(2x + y - 14) = 0$$

$$24x - 8y - 48 - 14x - 7y + 98 = 0$$

$$10x - 15y + 50 = 0$$

$$2x - 3y + 10 = 0 \quad 1 \text{ for correct equation}$$

$$(\text{or } y = \frac{2}{3}x + \frac{10}{3})$$

iii) $d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$

$$= \frac{|2 \times 1 + 4 - 14|}{\sqrt{2^2 + 1^2}}$$

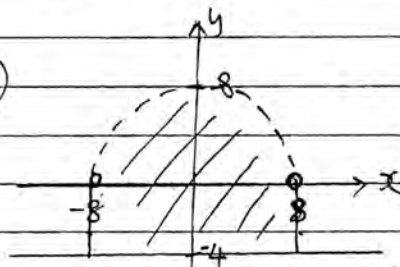
1 for working

$$= \frac{8}{\sqrt{5}} \text{ units}$$

1 for correct answer

$$(\frac{8\sqrt{5}}{5})$$

b)



1 for dotted semi circle

1 for filled vertical lines at ± 8

1 for filled line $y = -4$ and
correct region shaded

Q3(ii)

or: $3x - y - 6 = 0$

$$2x + y - 14 = 0$$

$$5x = 20$$

$$x = 4$$

$$y = 3x - 6 = 12 - 6 = 6$$

(4, 6) and (1, 4) ①

$$m = \frac{6 - 4}{4 - 1} = \frac{2}{3}$$

$$y - 4 = \frac{2}{3}(x - 1) \quad ①$$

$$3y - 12 = 2x - 2$$

$$\rightarrow 2x - 3y + 10 = 0 \quad ①$$

Q4 Logarithm and Indices

$$a) \log_3 15 = 2.46 \text{ (2 d.p.)} \quad 1$$

$$b) 64^{\frac{1}{3}} = a \\ \therefore a = 4 \quad 1$$

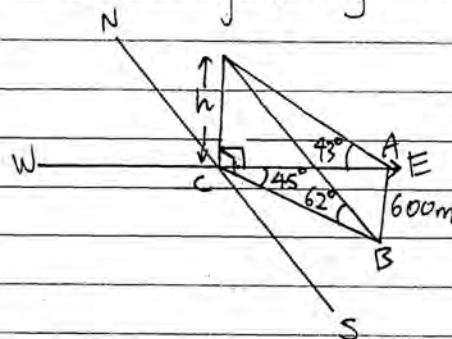
$$c) (\log_6 4 \times 9)^2 \\ = (\log_6 36)^2 \\ = 2^2 \\ = 4 \quad 1$$

$$d) b^{\log_b 7^2} \\ = 7^2 \\ = 49 \quad 1$$

$$e) \log_3 \frac{x+3}{x-3} = \log_3 3 \quad \swarrow 1 \quad \nwarrow 1 \\ \frac{x+3}{x-3} = 3 \quad /3 \\ x+3 = 3x-9 \\ 2x = 12 \\ x = 6 \quad 1$$

$$f) \log_2 2^{x+1} < \log_2 12 \quad 1 \\ x+1 < \frac{\log 12}{\log 2} \\ x < \frac{3.58-1}{2.58} \quad 1$$

Q5 3D Trigonometry



$$(i) \frac{h}{BC} = \tan 62^\circ \\ \therefore BC = \frac{h}{\tan 62^\circ} \quad 1$$

$$(ii) \frac{h}{CA} = \tan 43^\circ \\ \therefore CA = \frac{h}{\tan 43^\circ} \quad 1$$

(iii) using cosine rule:

$$AB^2 = \left(\frac{h}{\tan 62^\circ}\right)^2 + \left(\frac{h}{\tan 43^\circ}\right)^2 - 2 \times \frac{h}{\tan 62^\circ} \times \frac{h}{\tan 43^\circ} \cos 45^\circ \quad \begin{matrix} 2 \\ (-1 \text{ for each} \\ \text{wrong} \\ \text{working}) \end{matrix}$$

$$600^2 = h^2 \left(\frac{1}{(\tan 62^\circ)^2} + \frac{1}{(\tan 43^\circ)^2} \right) - \frac{2h^2 \cos 45^\circ}{\tan 62^\circ \tan 43^\circ}$$

$$= 1.43 h^2 - 0.81 h^2$$

$$360000 = 0.62 h^2$$

$$h^2 = 580645.16$$

$$h = 762 \text{ m} \rightarrow \text{not accurate} \quad 1 \\ 758.14$$